AI-BASED NETWORK INTRUSION DETECTION SYSTEM

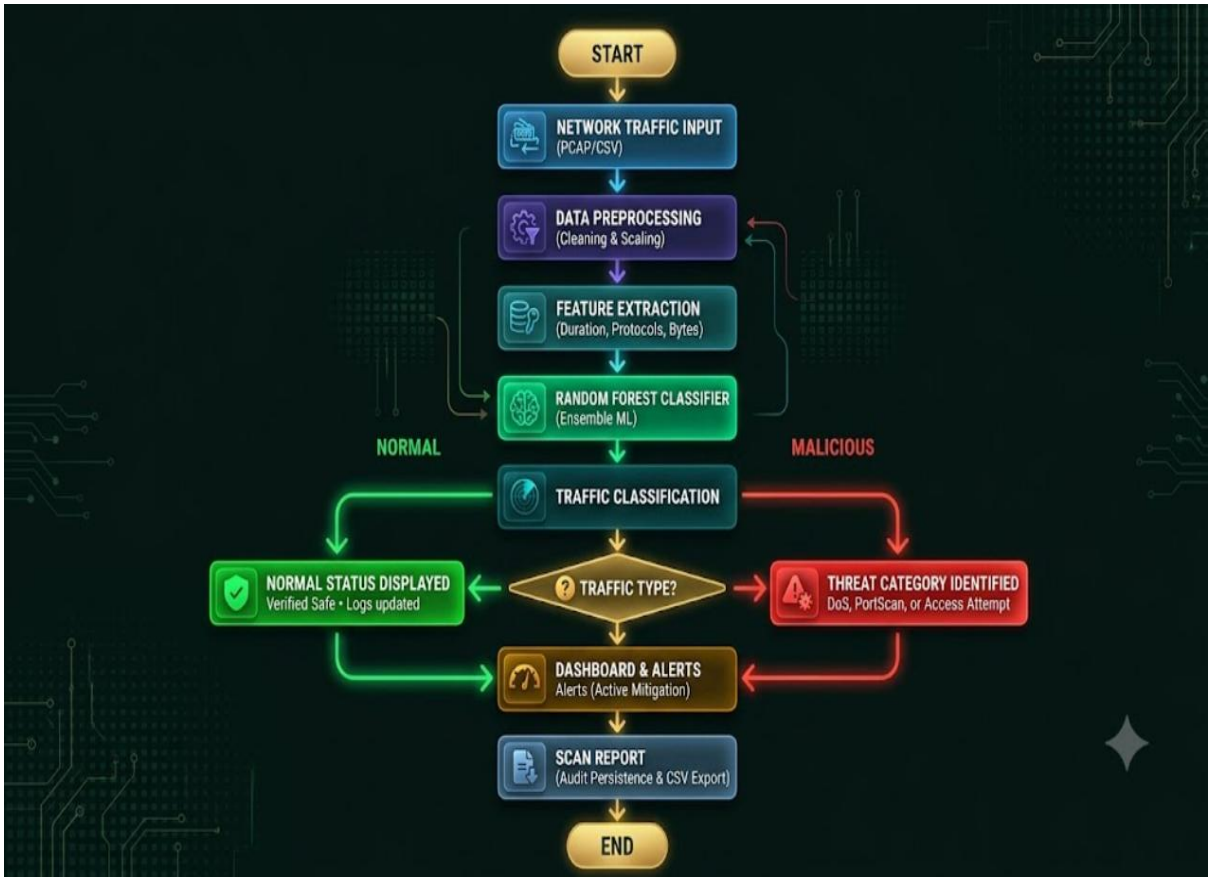
USER MANUAL

See the Threat. Understand the Risk. Defend the Network.

1. WELCOME TO INTELLIGENT CYBER DEFENSE

Every network tells a story. Hidden within its traffic, unusual patterns can reveal potential cyber threats. The AI-Based Network Intrusion Detection System (NIDS) transforms network traffic into meaningful security insights using machine learning, interactive visualizations, and intelligent threat classification.

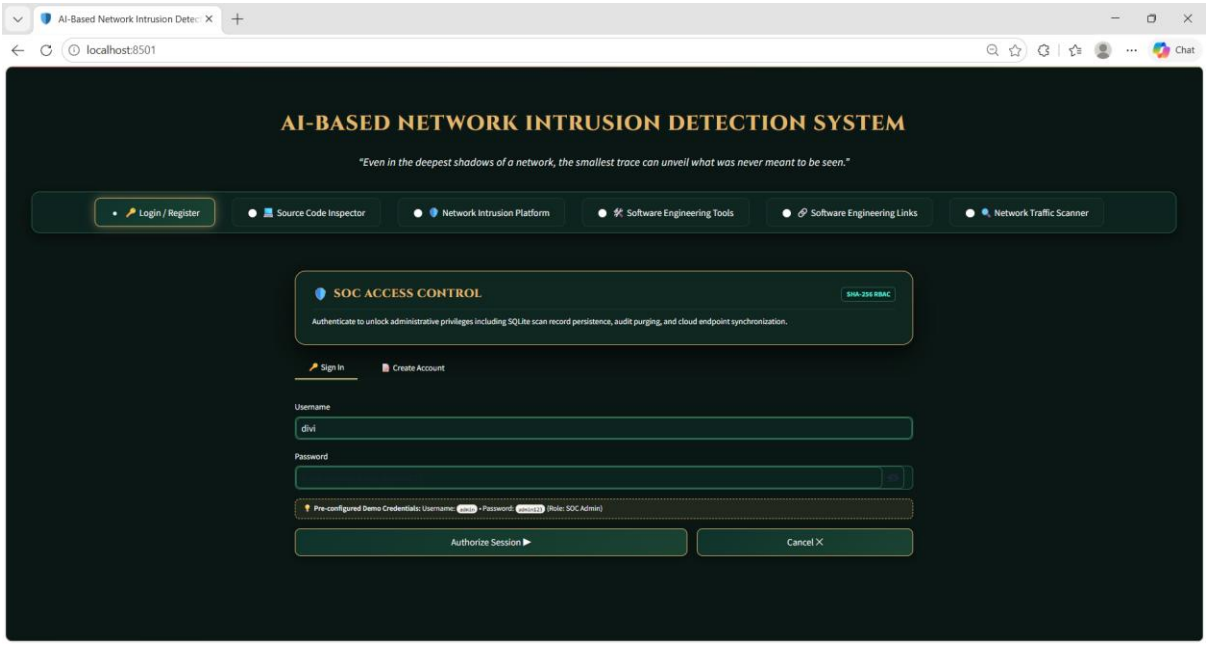
Powered by Python, Scikit-learn, Pandas, Plotly, and SQLite, the platform brings network analysis, intrusion detection, and security reporting together in one unified workstation.



2. EXPLORE YOUR SECURITY WORKSTATION

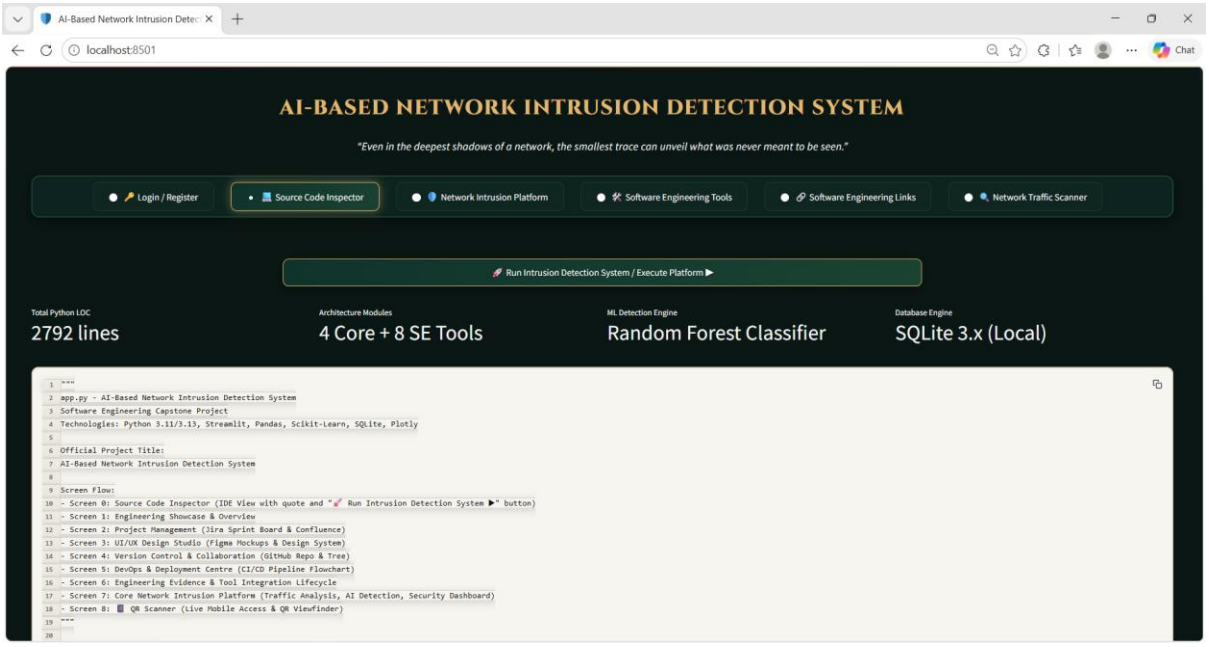
Login / Register

Access the application's authentication interface to sign in or create an account. Follow the on-screen instructions to enter your credentials and access the available features.



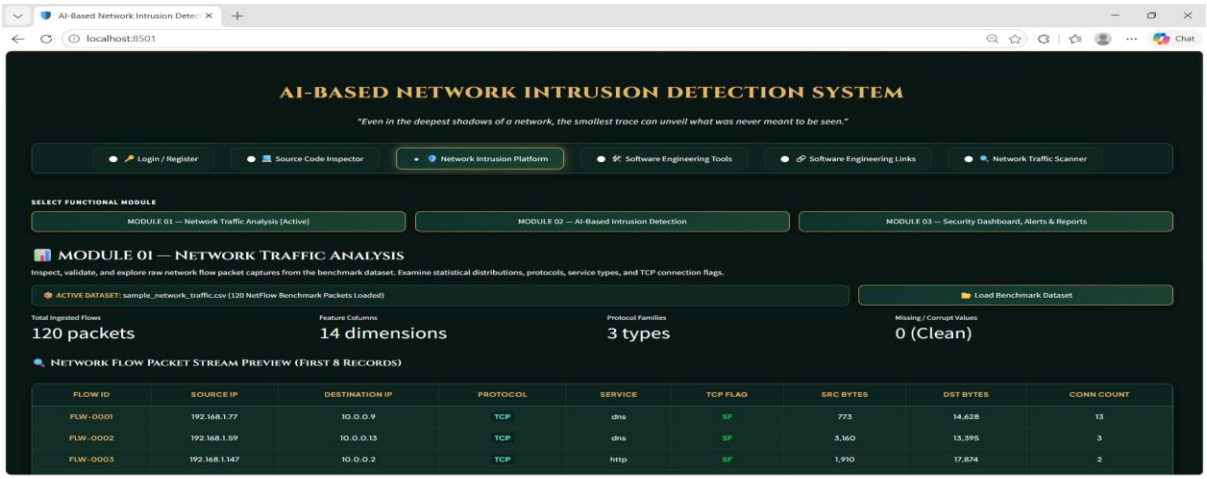
Source Code Inspector

Explore the project's code, architecture, and technical information. Use the Run Intrusion Detection System button when available to launch the platform's execution workflow.



🌓 Network Intrusion Platform

- Network Traffic Analysis: Examine datasets, protocols, services, and traffic patterns.
- AI-Based Intrusion Detection: Classify network flows and investigate potential threats such as DoS attacks, port scanning, and unauthorized access.
- Security Dashboard, Alerts & Reports: Review security statistics, threat distributions, incident alerts, and scan history.

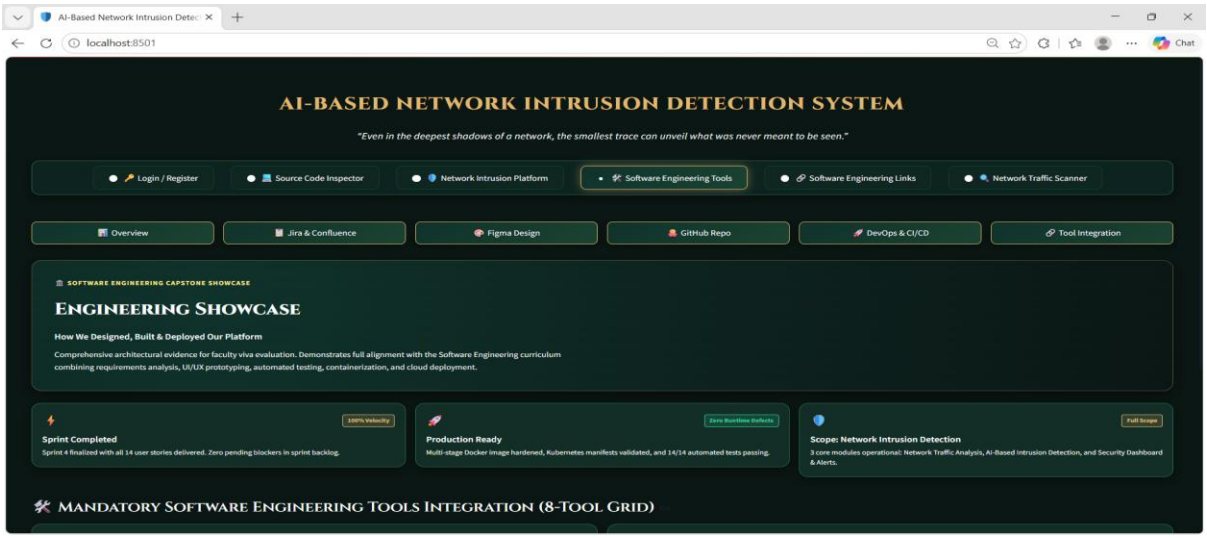


🛠 Software Engineering Tools

Explore eight tools supporting the software development lifecycle:

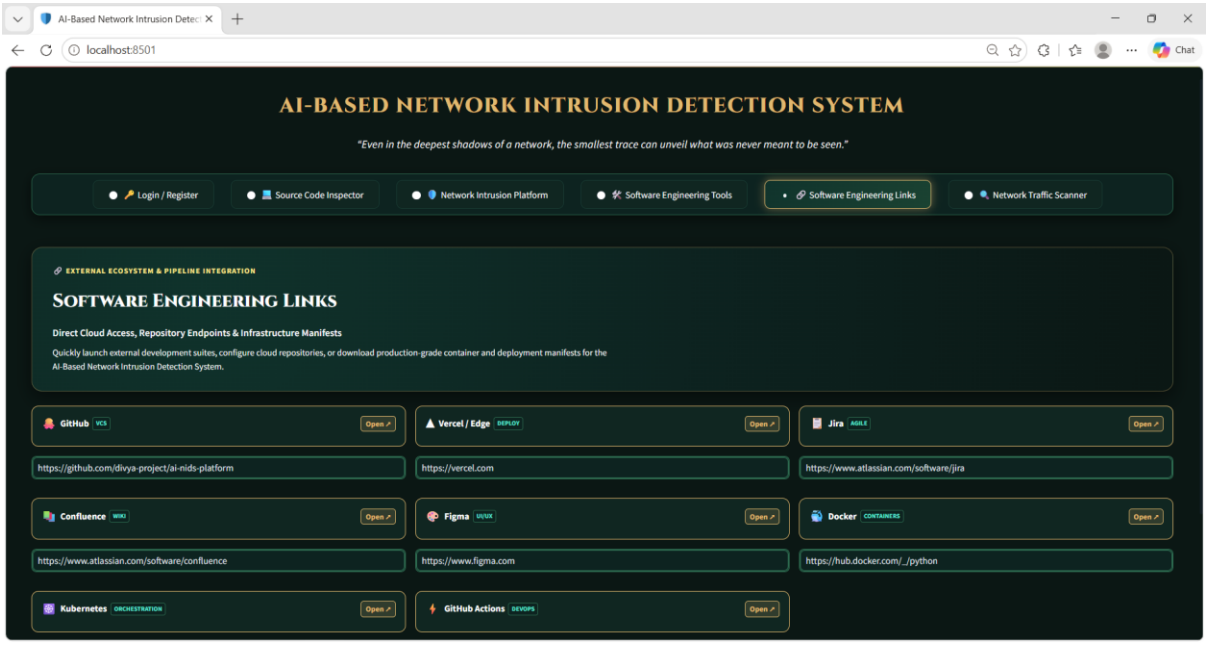
GitHub · Deployment · Jira · Confluence · Figma · Docker · Kubernetes · CI/CD

These tools support version control, project management, interface design, documentation, containerization, and deployment workflows.



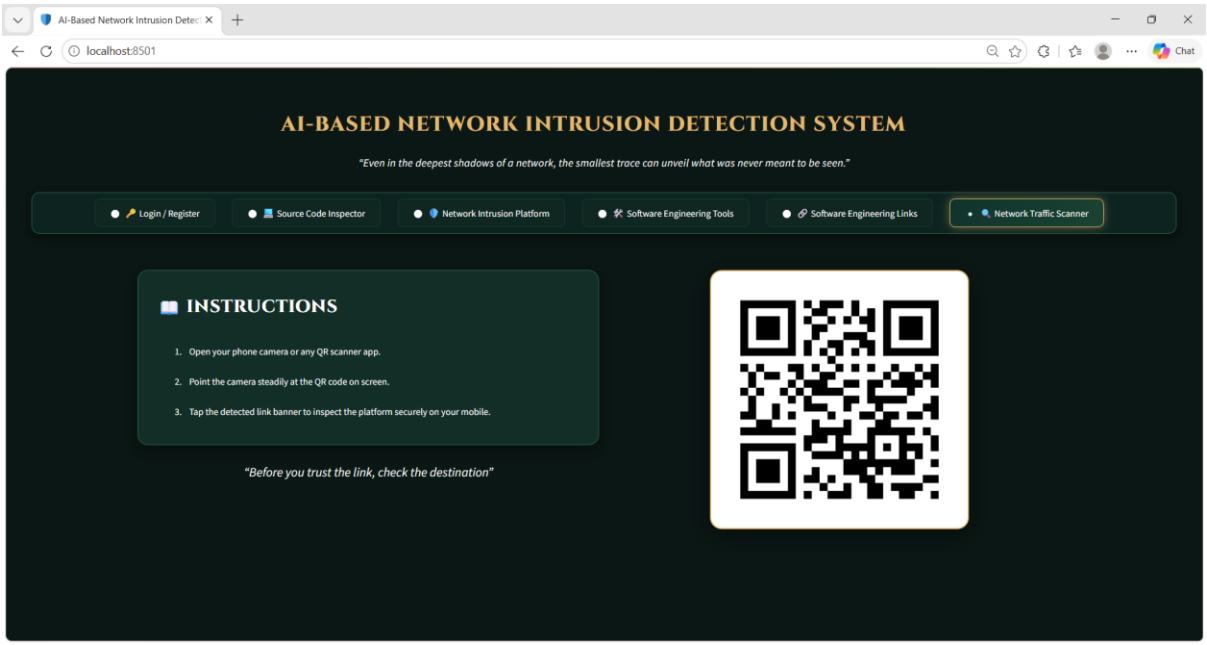
🔗 Software Engineering Links

Access the available external resources for development, collaboration, documentation, and deployment. Verify external websites before sharing credentials or sensitive information.



🔍 QR Scanner

Open the QR scanning interface and follow its on-screen instructions to scan or decode QR codes, where supported. Always verify a decoded link before opening it.



3. SCAN. DETECT. INVESTIGATE.



Follow this simple workflow to analyze network traffic:

STEP 1 — LOAD

Open Network Traffic Scanner and upload a supported CSV dataset, or select the benchmark dataset if available.

STEP 2 — ANALYZE

Click ANALYZE NETWORK TRAFFIC to begin processing the selected data.

STEP 3 — DETECT

Review the AI-generated predictions to distinguish normal traffic from potentially malicious activity.

STEP 4 — INVESTIGATE

Examine traffic statistics, threat categories, security alerts, and recommended mitigation actions.

STEP 5 — EXPORT

Use Download Scan Report to save the available scan results as a CSV file.

Note: The scanner analyzes uploaded or bundled datasets; it does not perform live network packet capture unless that functionality is separately implemented.

4. UNDERSTAND YOUR SECURITY INSIGHTS

The dashboard helps transform complex traffic records into understandable indicators.

- Total Flows: Number of analyzed network records.
- Normal Flows: Records classified as normal by the model.
- Malicious Flows: Records classified as potentially malicious.
- Anomaly Rate: Percentage of analyzed flows classified as malicious.
- Threat Classification: Predicted attack categories for further investigation.
- Security Alerts: Available incident details and recommended responses.

The Random Forest model uses learned network traffic patterns to generate predictions. Accuracy, precision, recall, and F1-Score help evaluate its performance. Any displayed benchmark metrics should reflect verified evaluation results.

Remember: An AI prediction is a security indicator, not absolute proof of an attack. Investigate suspicious findings before taking action.

5. QUICK TROUBLESHOOTING

Dataset not loading? Check the CSV format and required columns.

Predictions failing? Verify the input features and model availability.

Charts appearing empty? Confirm that the dataset contains usable records and the required fields.

Application not opening? Check the installed dependencies, configuration, and terminal error messages.

6. DEFEND RESPONSIBLY

Use authorized network data, protect account credentials, safeguard exported reports, and verify mitigation commands before execution. Test security changes carefully and follow your organization's policies.

NIDS supports cybersecurity investigation; it does not replace professional security monitoring or incident response.

“INTELLIGENCE IN EVERY FLOW, VISIBILITY IN EVERY THREAT”